Plots Merge

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1 What are we plotting?

We aim to segment the data into 4 groups:

1. Plan had above/below median alpha
2. Plan was terminated or not-terminated

Within a given year, each group assigned to a beneficiary will be unique. However, a beneficiary may change plans across years for various reasons: If their plan terminates they would be forced into a new plan, or if they move they would chanage plans by definition (since we have determined that the "same" plan is different in different counties).

Even if their plan does not terminate and they do not move, they may appear with a new plan the next year. Hence this definition is not unique across years. Since we want to compare a ***cohort*** of people, when we make the plots we fix a year, call it t, and we segment the sample using plan alphas and terminations from year $t-1$ using a single year of data. Hence we have a single data point per beneficiary which indicates which of the 4 groups they belong to:

Group 1: Above median, non-terminated

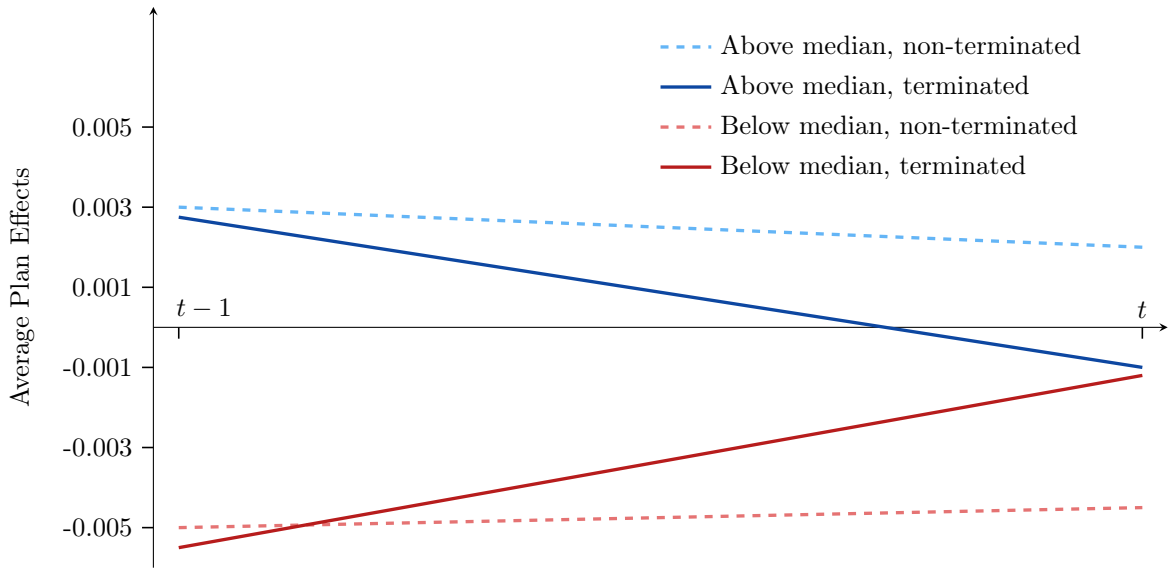
Group 2: Above median, terminated

Group 3: Below median, non-terminated

Group 4: Below median, terminated

We merge this single year of data into every other year, giving us a way to track the cohort over time that is consistent. The resulting plot will look like this:

Figure 1.1: Average Plan Effects Over Time



With each line representing the average plan effect in the indicated year using the cohort groups for year t, which were defined using $t-1$ variables.

2 Plots version 1 ("old")

For a given beneficiary, we have the following:

Table 2.1: Example of Data for Beneficiary 1

bene	year	plan	ma	term	above median
1	2006	1	1	0	0
1	2007	1	1	0	1
1	2008	2	0	0	1
1	2009	2	0	0	0
1	2010	1	1	1	0
1	2011	2	0	0	1

For simplicity in this example, we write "plan" when we mean county-plan, and we will consider a single year of data: 2011, which is the year with the most terminations. Since 2011 is the reference year, that is, $t = 2011$, the plan we care about is the $t - 1 = 2010$ plan, which is plan #1. Note that it does not matter that the 2011 plan is no longer an MA plan, and that the beneficiary was not enrolled in an MA plan in 2009. We only care about the $t - 1 = 2010$ data to assign a group to this beneficiary, which in this case is "Terminated, Below Median": Group 4.

In general, for reference year 2011 we have 2010 data

Table 2.2: Assigning Groups for Reference Year 2011

bene	year	plan	ma	term	above median	group
1	2010	1	1	1	0	4
2	2010	2	0	0	.	.
3	2010	3	1	1	0	4
4	2010	4	0	1	.	.
5	2010	5	1	1	0	4
6	2010	6	1	1	1	2
7	2010	7	1	0	0	3
8	2010	8	1	0	1	1

Note that non-MA plans are dropped, regardless of terminated status. So they would neither be assigned an above/below median indicator nor a group for the plots. To produce the $t - 1, t$ time series plots, we would keep only `bene` and `group` from this data and merge by `bene` onto the main data, as illustrated in the following table (note that group is time-invariant, even as other variables change):

Table 2.3: Merge in Group Assignments for Reference Year 2011

bene	year	plan	ma	term	above median	group
1	2010	1	1	1	0	4
2	2010	2	0	0	.	.
3	2010	3	1	1	0	4
4	2010	4	0	1	.	.
5	2010	5	1	1	0	4
6	2010	6	1	1	1	2
7	2010	7	1	0	0	3
8	2010	8	1	0	1	1
1	2011	2	0	0	.	4
2	2011	2	0	0	.	.
3	2011	8	1	0	1	4
4	2011	8	1	0	1	.
5	2011	2	0	0	.	4
6	2011	7	1	0	0	2
7	2011	7	1	0	0	3
8	2011	8	1	0	1	1

3 Plots version 2 (current)

We always want to use $t-1$ variables to compute the groups by beneficiary. However, for the IV regressions, we use $t-1$ variables at year t: That is, we use lag variables in our main specification and to construct our instrument. Hence we wanted to make a version of the plots that was consistent with this paradigm. This is achievable by replicating Table 3.1 using $t-1$ (lag) variables at year t:

Table 3.1: Assigning Groups for Reference Year 2011

bene	year	lag plan	lag ma	lag term	lag above median	group
1	2011	1	1	1	0	4
2	2011	2	0	0	.	.
3	2011	3	1	1	0	4
4	2011	4	0	1	.	.
5	2011	5	1	1	0	4
6	2011	6	1	1	1	2
7	2011	7	1	0	0	3
8	2011	8	1	0	1	1

Note that the groupings are identical in this case, but the year is now 2011 and we are instead using lag versions of the same variables. From here it looks tempting to proceed just as in Section 2: Keep `bene` and `group` then merge by `bene` onto the main data. However, we need to account for deaths: The above only defines the groups for *surviving* beneficiaries. We actually need to insert two additional steps:

1. Rename `lag plan` to `plan` and replace `year` with `year - 1`.

2. Merge by `plan` (which has the `lag plan` data) and year into the main data.

This would assign each of the 4 groups in year $t-1$ to all beneficiaries enrolled in the above plans regardless of whether they are dead in year t or not. From here we *can* proceed as in Section 2, but it should be clear that this is ***not equivalent*** to version 1:

1. If we define above/below median at year t using plan effects from year $t-1$, we are weighting by the number of *surviving* beneficiaries from the plan, as opposed to by all beneficiaries who were enrolled.
2. If every beneficiary in a terminated plan died, the plan would get dropped from the plots altogether.
3. Since the regressions are meant to be "stacked," that is, taken year-by-year, any selection criteria that is applied would be carried backwards, imposing a stronger selection criteria than intended for the plots. For example, if we requite `lag ma` to equal 1, then we would exclude in year $t-1$ every beneficiary who was not in an MA plan in year $t-2$. Since we merge the groups by plan in year $t-1$ to assign them to dead people, we are effectively imposing an "always MA" requirement as opposed to a "lag MA" restriction.

   Note that as long as we use the `ifcommand()` option to apply lag restrictions (like `lag ma == 1`) this last point should not be a problem.